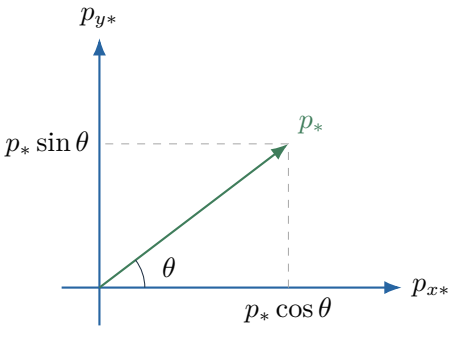


洛伦兹变换

沿 $+x$，速率 $v$

$$
\begin{aligned}
E &= \gamma(E_* + v p_{x*}) \\
p_x &= \gamma(p_{x*} + v E_*/c^2) \\
p_y &= p_{y*}
\end{aligned}
$$